Roll No.....

ODD END SEMESTER EXAMINATION DECEMBER – 2024

Name of the Program: M.Tech.(CSE)
Name of the Course: Mobile Computing

Semester: 1ST
Course Code: MCS 102

Time: 03 Hours
Note:-

Maximum Marks: 100

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	(20 marks)	CO1
a)	What is meant by handoff interval and handoff region? Explain their usefulness with appropriate diagram.	
b)	Explain HLR and VLR concept for location management in mobile communication. What is role of CoA? How is it assigned?	
c)	The Computer networking architecture consist of seven layers, as given in the ISO reference model. In your opinion, in which layer (s) should mobility be incorporated and why?	
Q2	(20 marks)	CO2
a)	Adaptive Clustering for Mobile Wireless Networks is essential. Discuss the advantages pored for such networks and also discuss the multi-hopping concepts over conventional cellular networks?	
b)	Define design goals of Coda? Discuss about Callback promise and Callback break in Coda File System.	
c)	What are the two basic important ideas on which Rover toolkit works. Explain the working mechanism of this toolkit.	
Q3	(20 marks)	CO3
a)	What is pointer forwarding strategy? What are required schemes, which are used to reduce signalling traffic?	
b)	Show clearly how IP-in-IP tunneling is used in mobile IP.	
c)	What is the need for energy efficient indexing and how can we manage broadcast data on air using indexing method?	
Q4	(20 marks)	CO4
a)	What do you understand by Mobile agent? Describe the model for mobile agent based computing. Also discuss the possible ways for providing security to mobile agents.	
b)	How mobile agents are helpful in the parallel computing and data collection.	
c)	What is Computer forensic? Discuss about the Cyber bulling. In order to cyber security Intrusion Detection became an important aspect. Therefore, Describe the key methods for Intrusion Detection.	
Q5	(20 marks)	CO5
a)	Classify the Adhoc network? What are the main functions and the design issues of medium access control layer in Mobile Adhoc Networks?	
b)	List out the problems in message routing in Adhoc wireless networks? Distinguish between proactive and reactive routing protocols. Explain one proactive routing protocol.	
c)	Indicate the purpose of 'Echo' and 'Hello' packet in the routing. Compare and contrast the routing mechanisms in mobile IP and AODV.	